

Relation Algebra

§6. Recursive Programs

§6.1. $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$

(6.1a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

(6.1b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.2 and every chapter after it uses
$\text{Monotonic}(\varphi) \implies \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski: the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

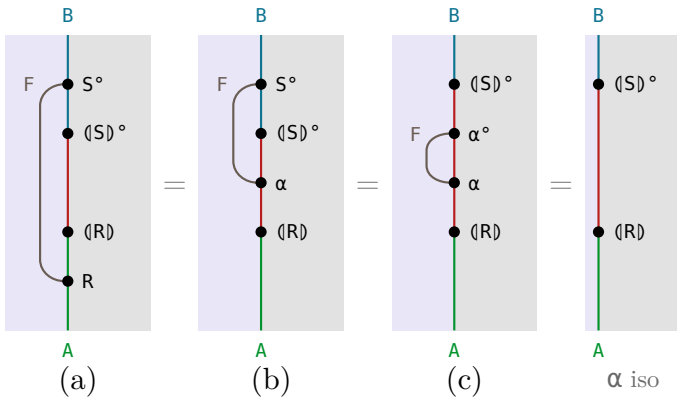
§6.2. $(S)^\circ (R) = (\mu X : S \circ F(X) R)$

$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : S^\circ F(X)R) \quad (6.2a)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



(a) the body at $(\mathbb{S})^\circ(\mathbb{R})$

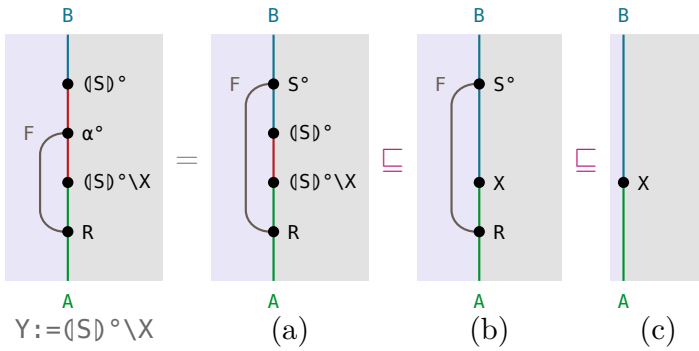
$F(RS) = F(R)F(S) \triangleq \text{relator}$

(b) $F((\mathbb{R}))R = \alpha(\mathbb{R}) \triangleq \text{reduce}$

(c) $(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ) \triangleq \text{reduce, relator-laws}$

$$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation



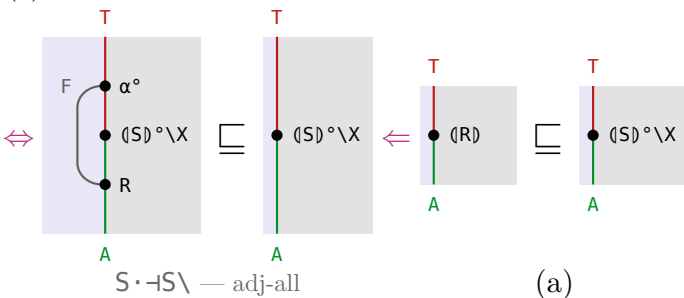
$Y := (\mathbb{S})^\circ \setminus X$

(a) $(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ)$

$F(RS) = F(R)F(S) \triangleq \text{relator}$

(b) $(\mathbb{S})^\circ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \triangleq \text{adj-all}$

(c) $S^\circ F(X)R \sqsubseteq X$



$S \cdot \dashv S \setminus \triangleq \text{adj-all}$

(a)

(a) $(\mathbb{R}) = (\mu X : \alpha^\circ F(X)R) \triangleq \text{reduce, (6.1b), adj-all}$